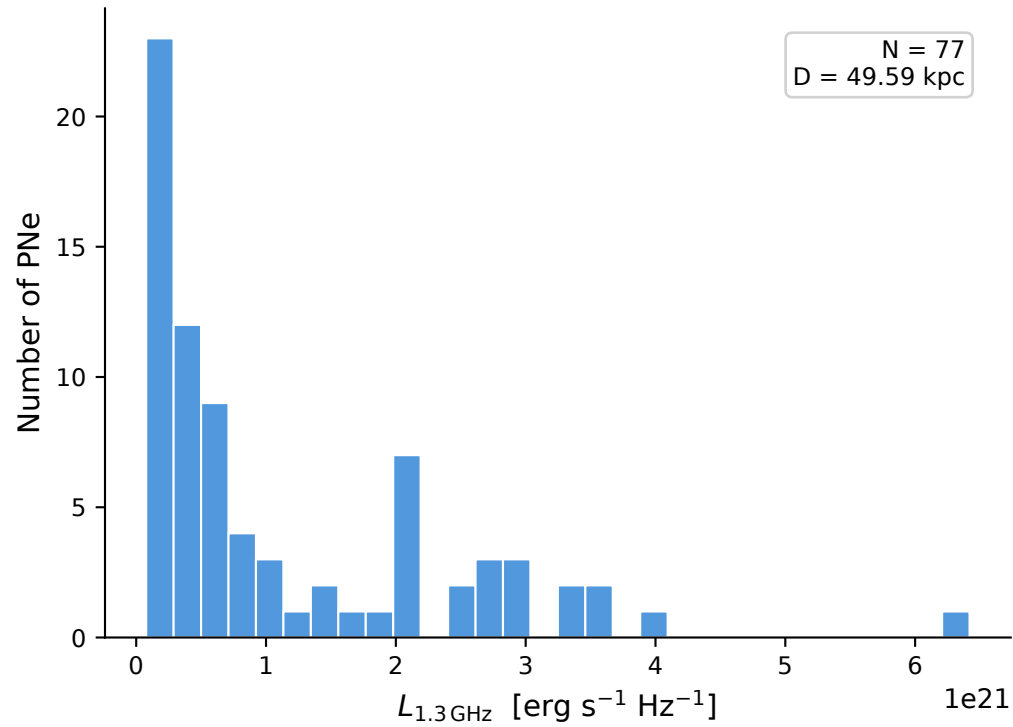
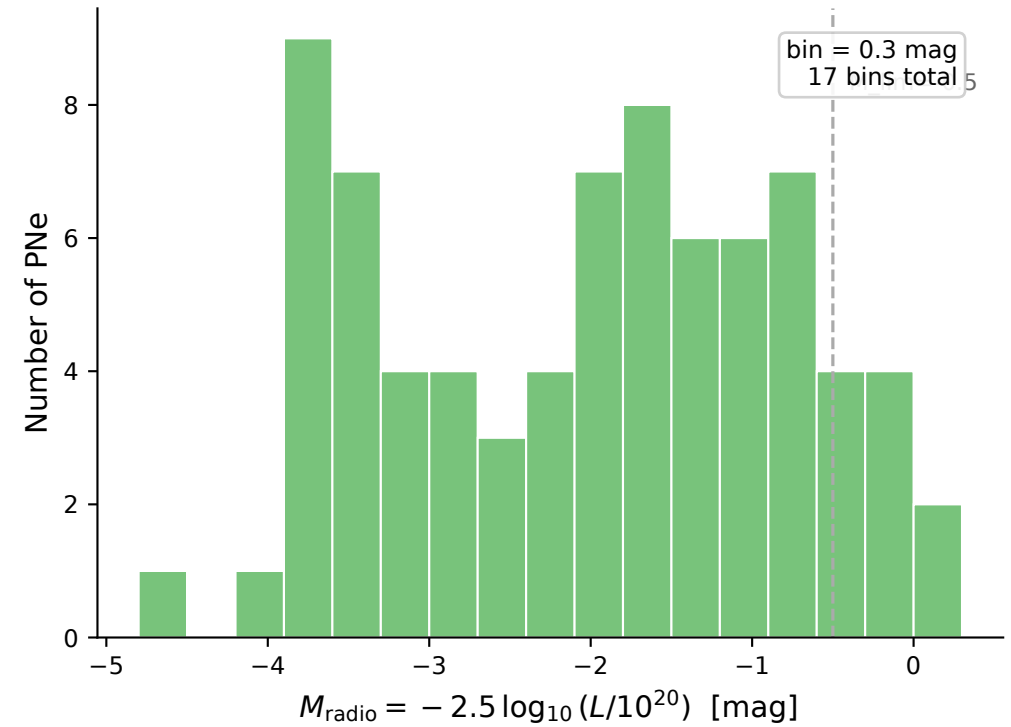


# Data Preparation: From Luminosity to Fit-Ready PNLF

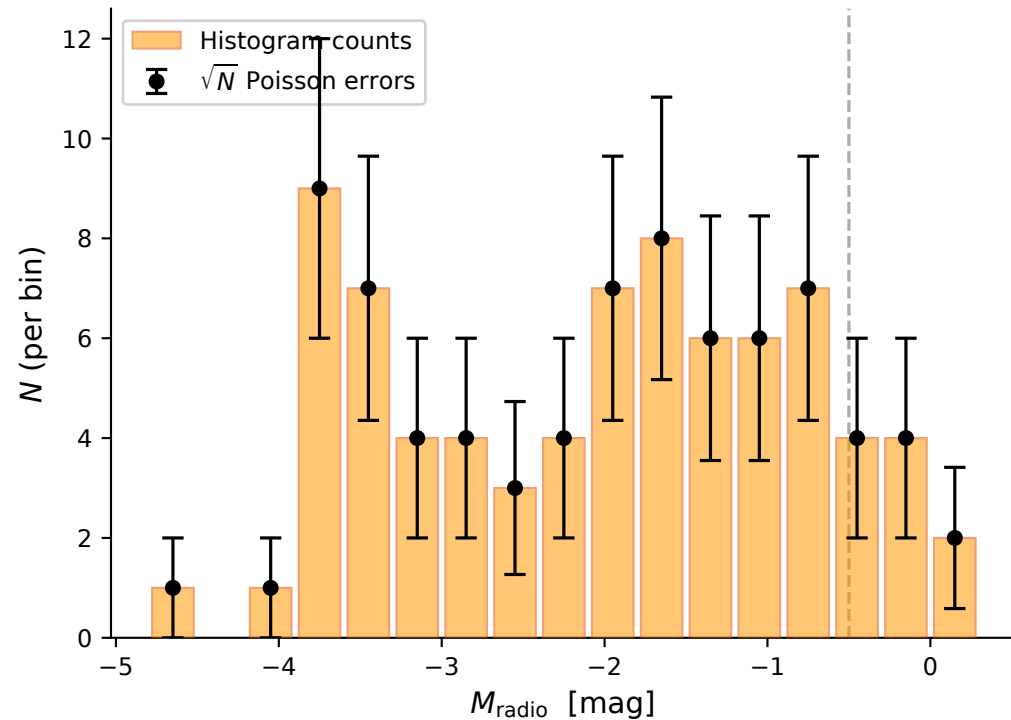
**(a) Luminosity distribution**



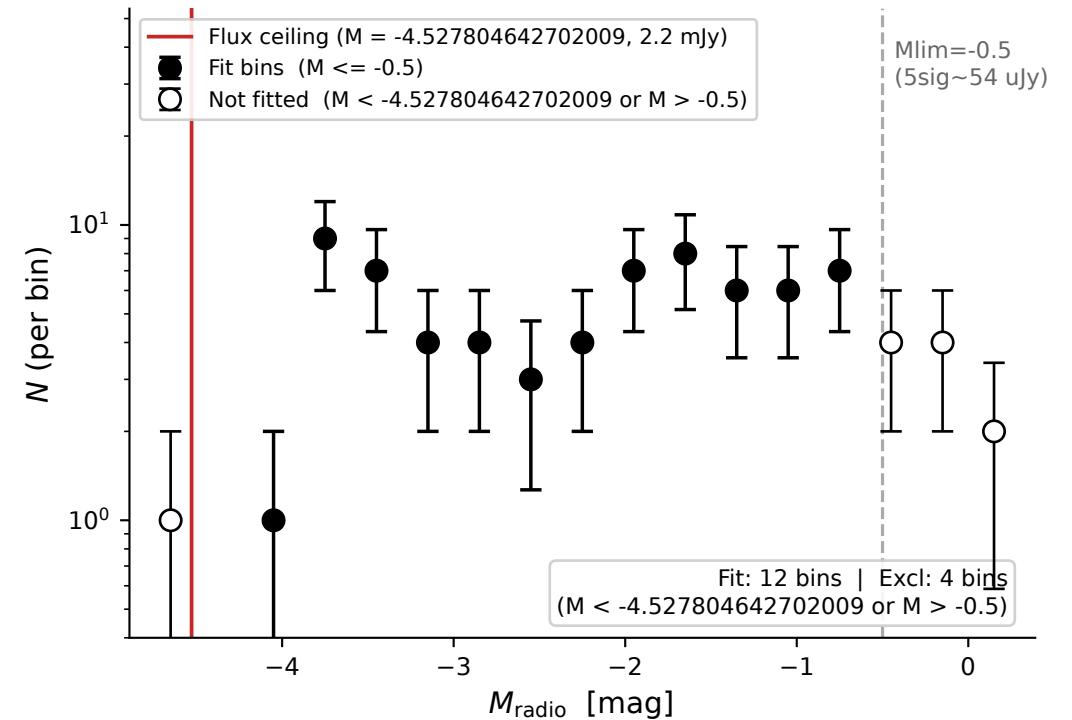
**(b) Magnitude distribution**

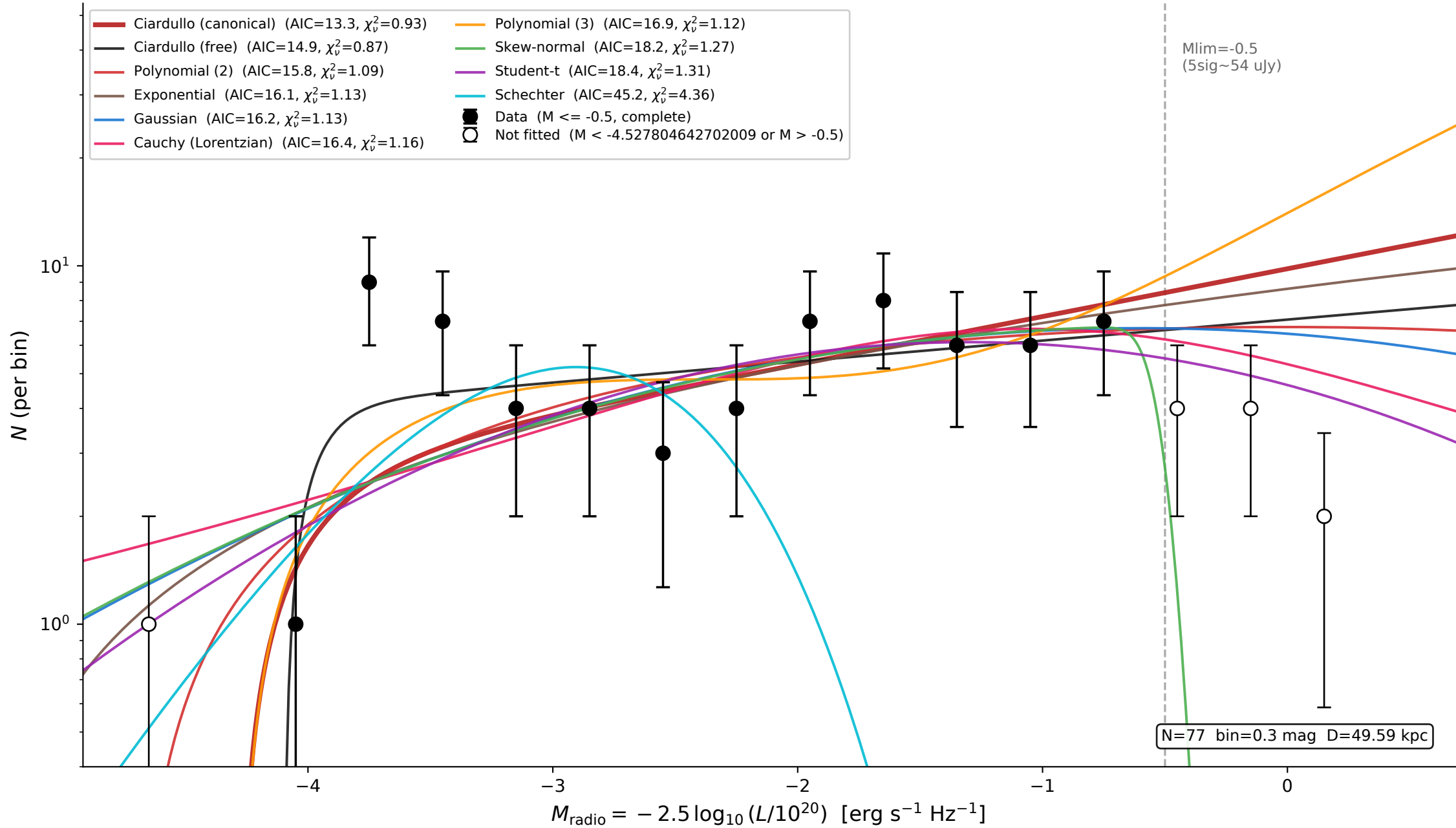


**(c) Binned counts with Poisson uncertainties**

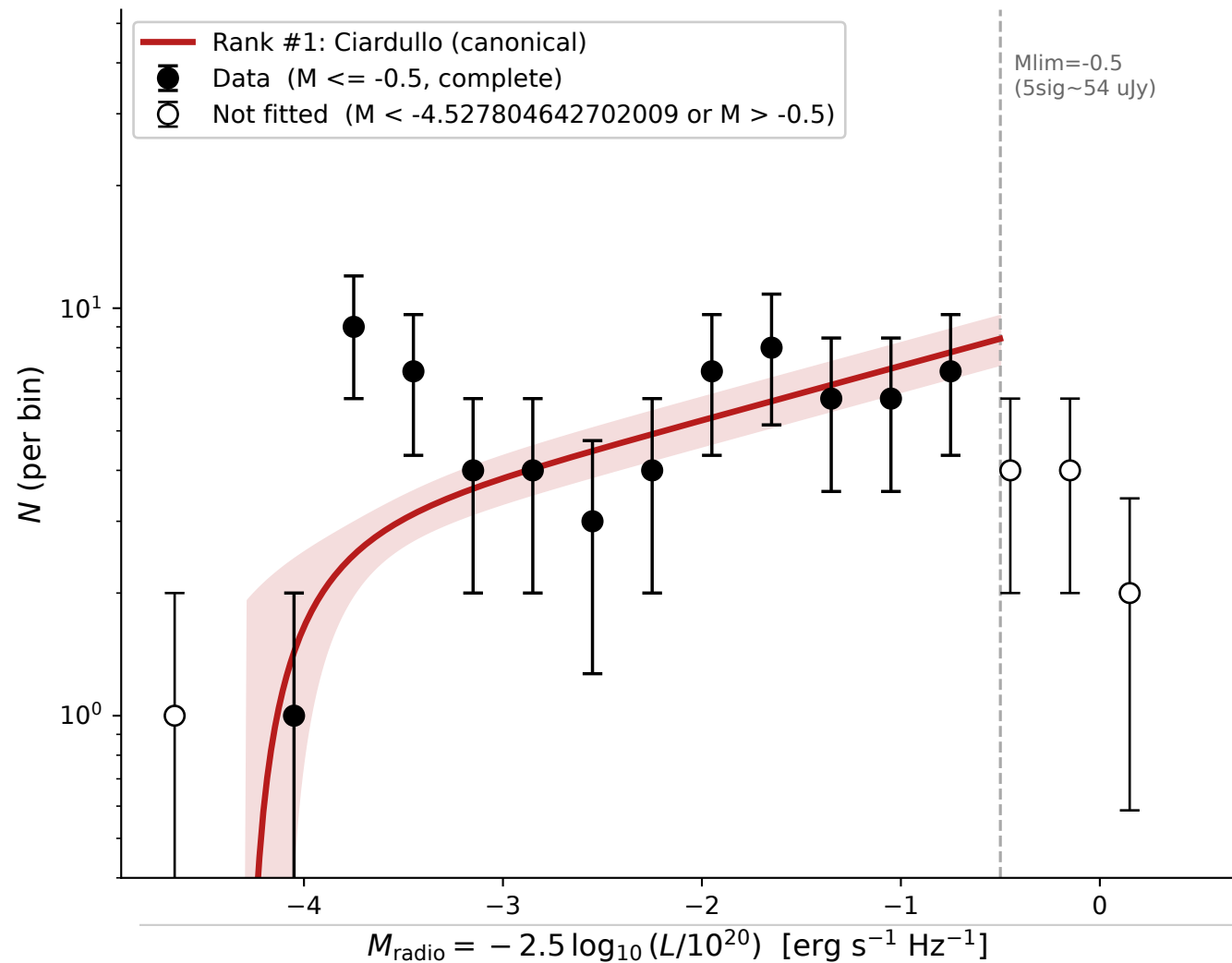


**(d) Fit-ready data: complete vs excluded**





## Rank #1: Ciardullo (canonical)



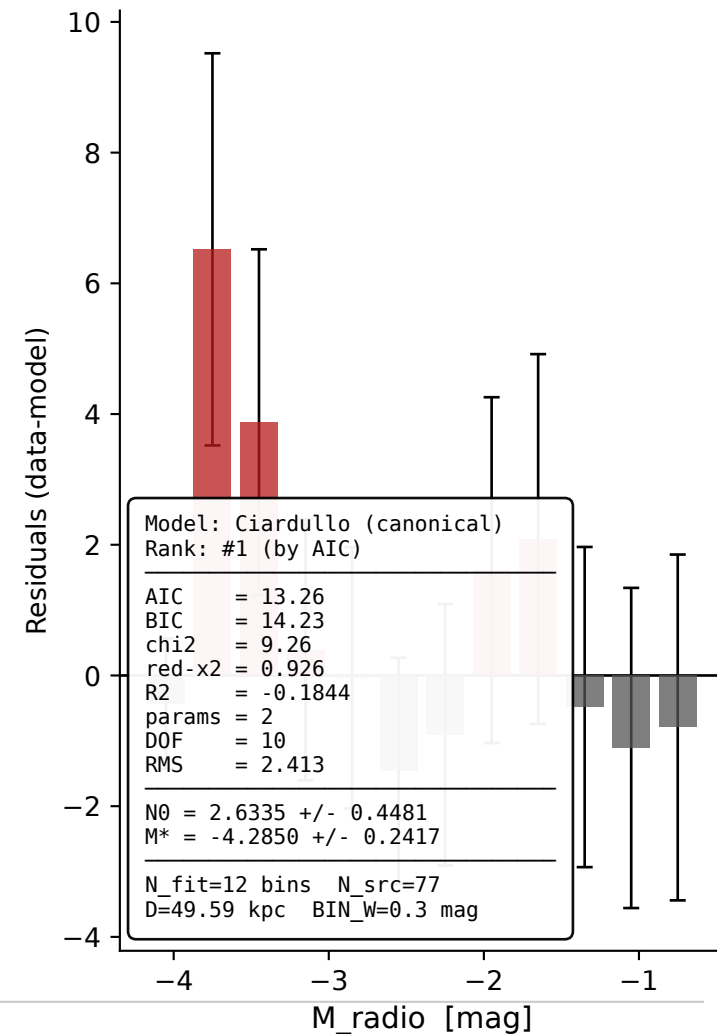
### General model equation:

$N(M) = N_0 * \exp(0.307 * (M - M^*)) * (1 - \exp(3.0 * (M^* - M)))$   
 $\alpha = 0.307$ ,  $\beta = 3.0$  (FIXED – Ciardullo 1989 optical values)  
 Only  $N_0$  and  $M^*$  are free ( $k=2$ ) – most parsimonious Ciardullo form

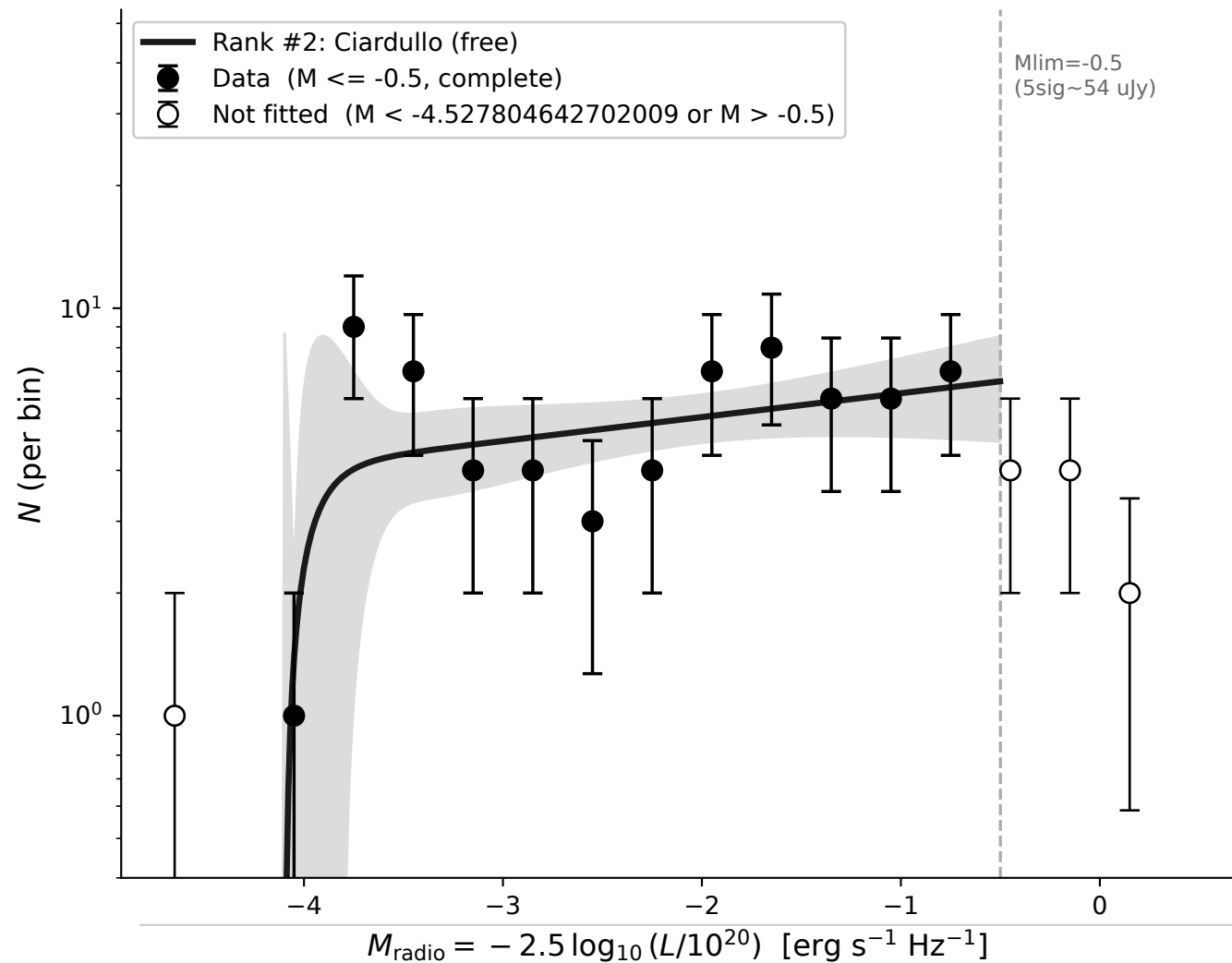
### Fitted equation (parameter values substituted):

$N(M) = 2.63 * \exp(0.307 * (M - (-4.285))) * (1 - \exp(3.0 * (-4.285 - M)))$   
 $N_0 = 2.63$   $M^* = -4.285$   $\alpha = 0.307$  (fixed)  $\beta = 3.0$  (fixed)  
 $L^* = 10^{21.714} \text{ erg/s/Hz}$

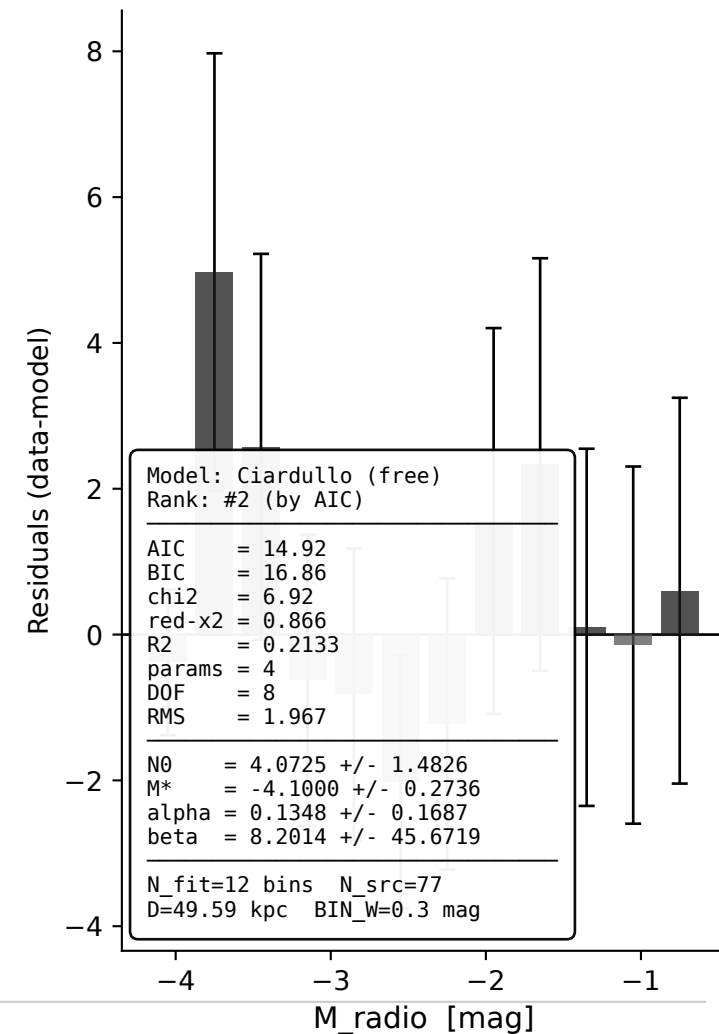
## Residuals



## Rank #2: Ciardullo (free)



## Residuals



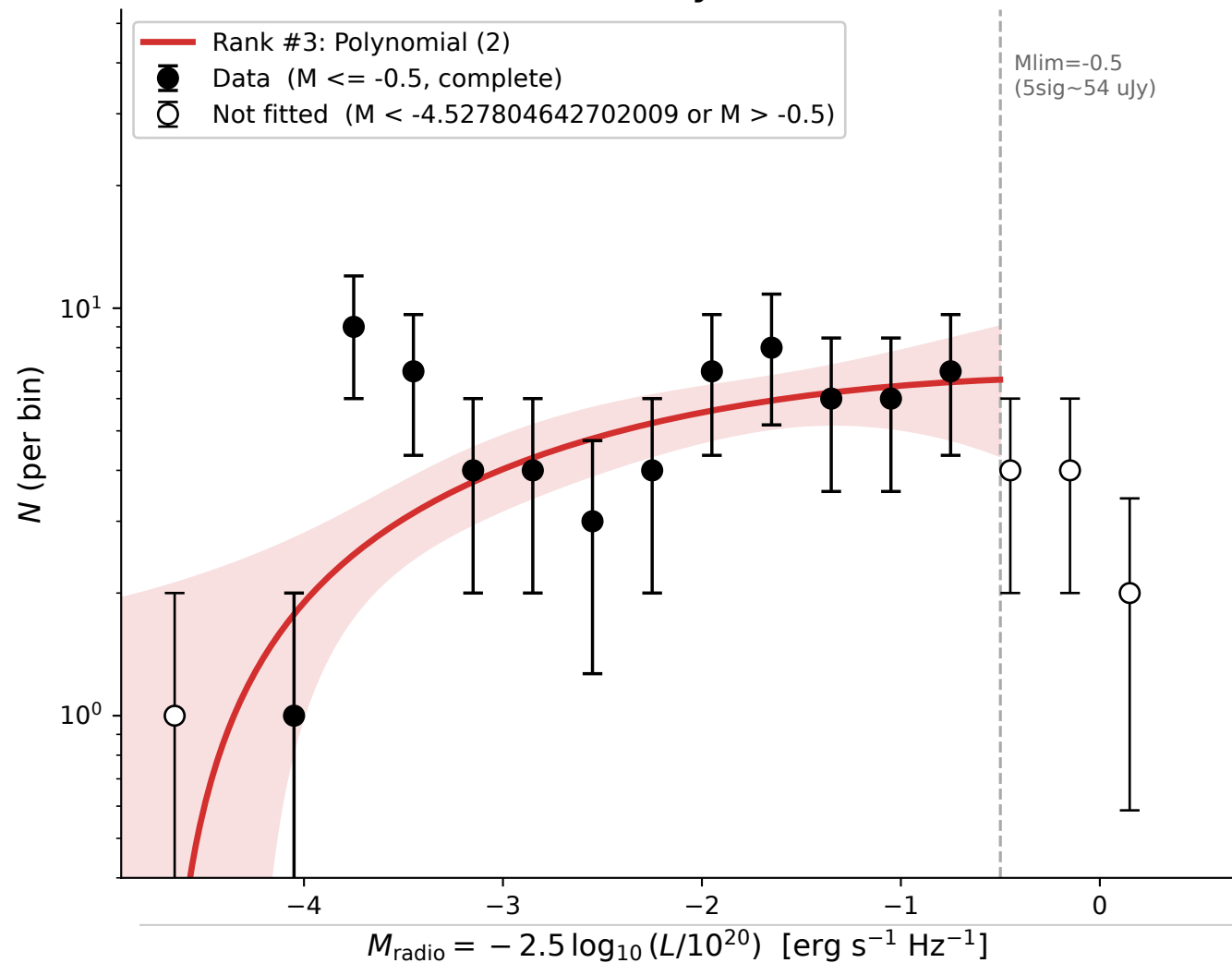
### General model equation:

$$N(M) = N_0 * \exp(\alpha * (M - M^*)) * (1 - \exp(\beta * (M^* - M)))$$
  
ALL parameters free:  $N_0$ ,  $M^*$ ,  $\alpha$  (faint-end slope),  $\beta$  (bright cutoff)  
Ciardullo (1989) form – radio calibration, not fixed to optical values

### Fitted equation (parameter values substituted):

$$N(M) = 4.07 * \exp(0.135 * (M - (-4.100))) * (1 - \exp(8.201 * (-4.100 - M)))$$
  
 $N_0=4.07$   $M^*=-4.100$   $\alpha=0.135$   $\beta=8.201$   
 $L^* = 10^{21.640} \text{ erg/s/Hz}$

# Rank #3: Polynomial (2)



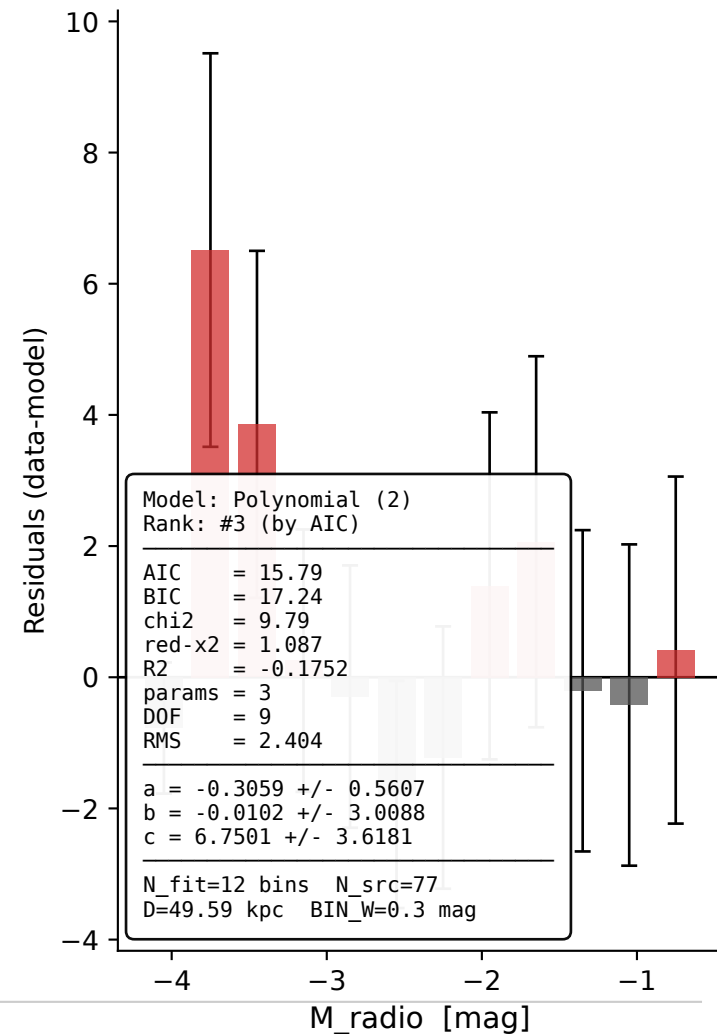
General model equation:

$$N(M) = a \cdot M^2 + b \cdot M + c$$

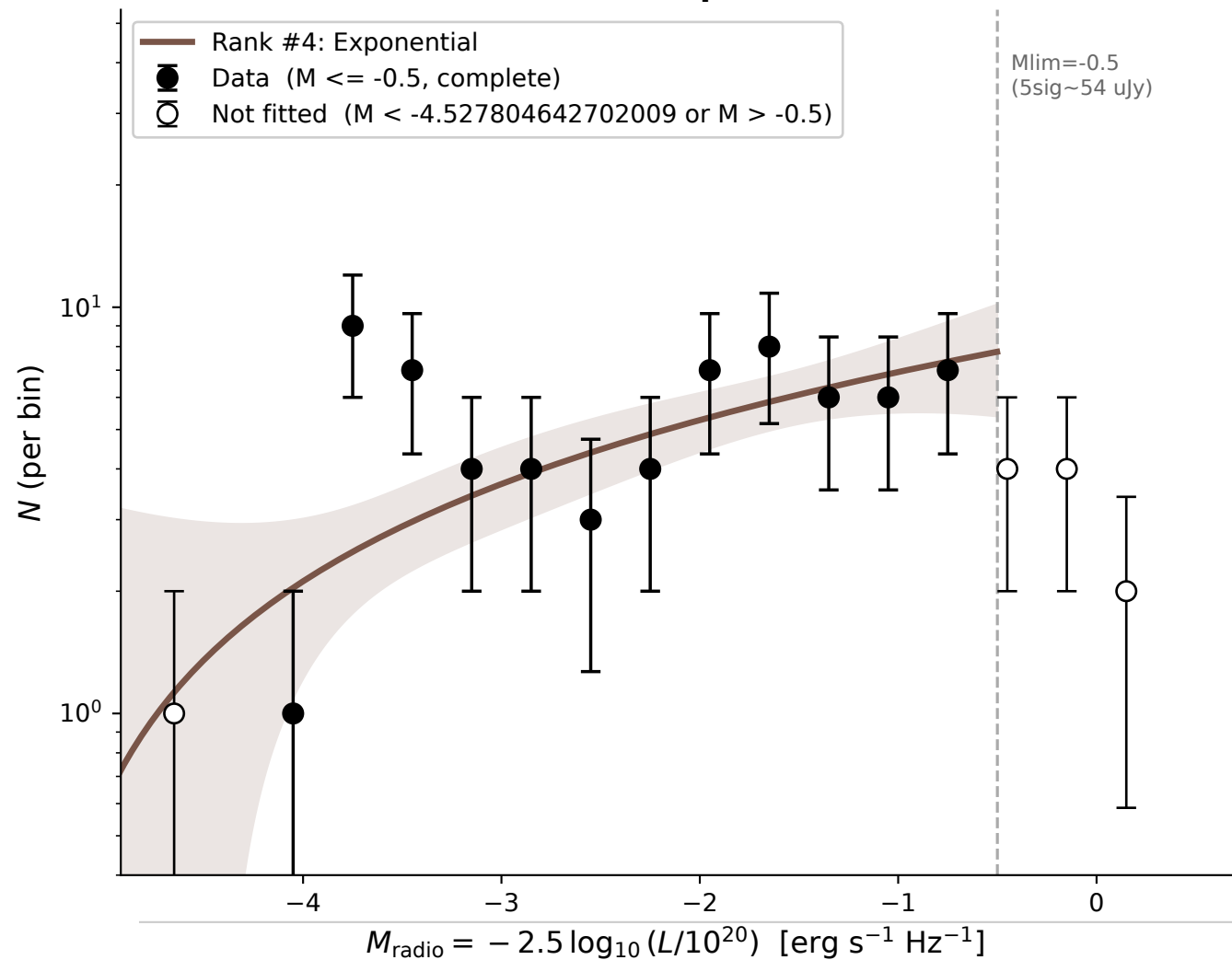
Fitted equation (parameter values substituted):

$$N(M) = -0.3059 \cdot M^2 - 0.0102 \cdot M + 6.7501$$

## Residuals



## Rank #4: Exponential



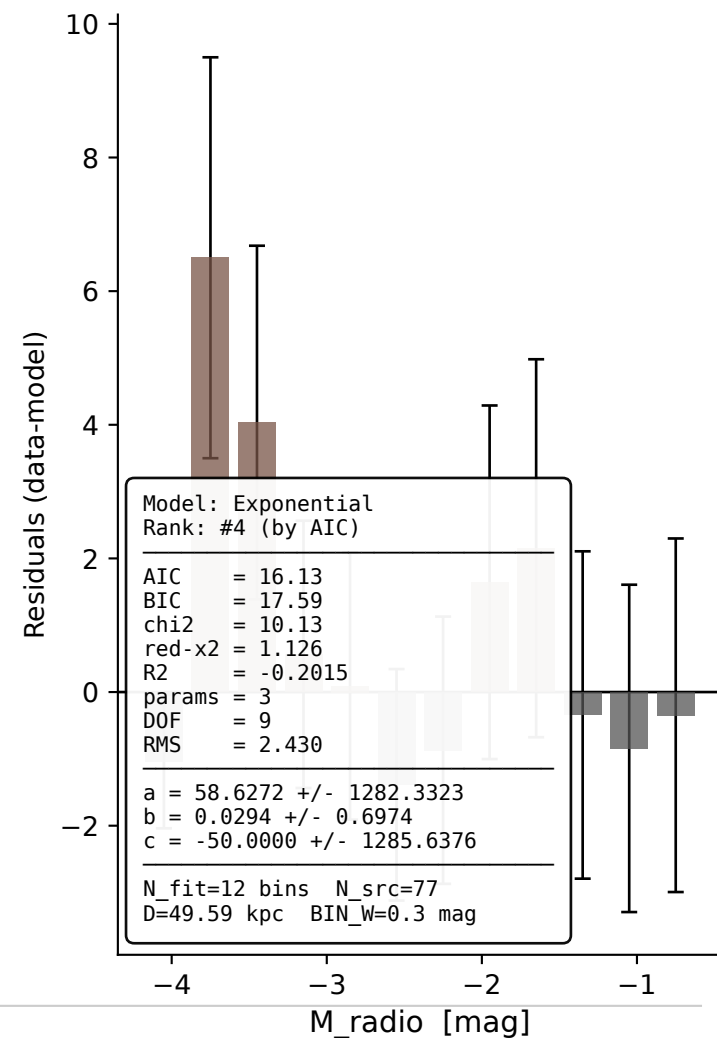
### General model equation:

$$N(M) = a * \exp(b * M) + c$$
 Simple exponential rise toward faint magnitudes  
 a = amplitude, b = rate (>0 = rising toward faint), c = offset

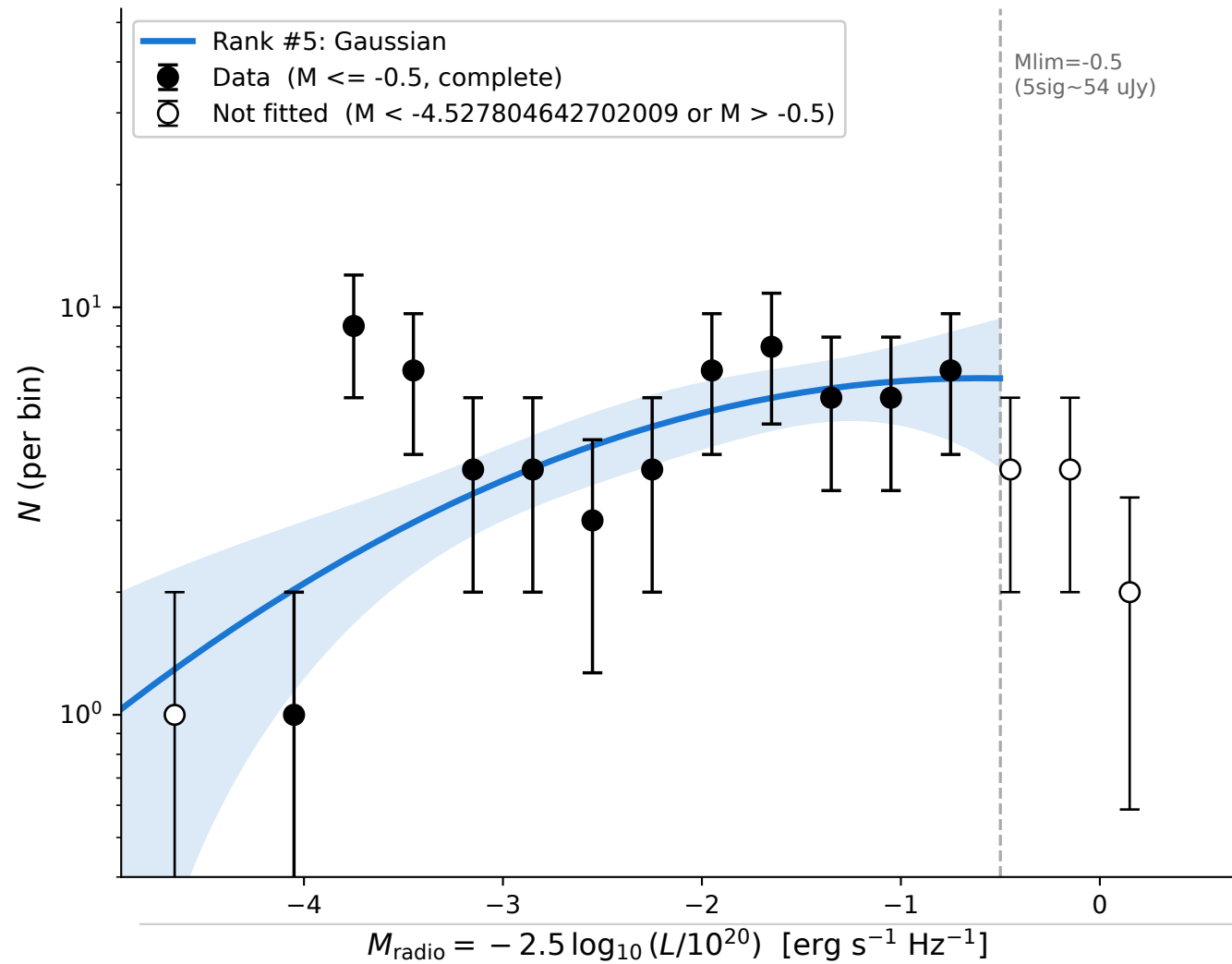
### Fitted equation (parameter values substituted):

$$N(M) = 58.6272 * \exp(0.0294 * M) - 50.0000$$

## Residuals



## Rank #5: Gaussian



General model equation:

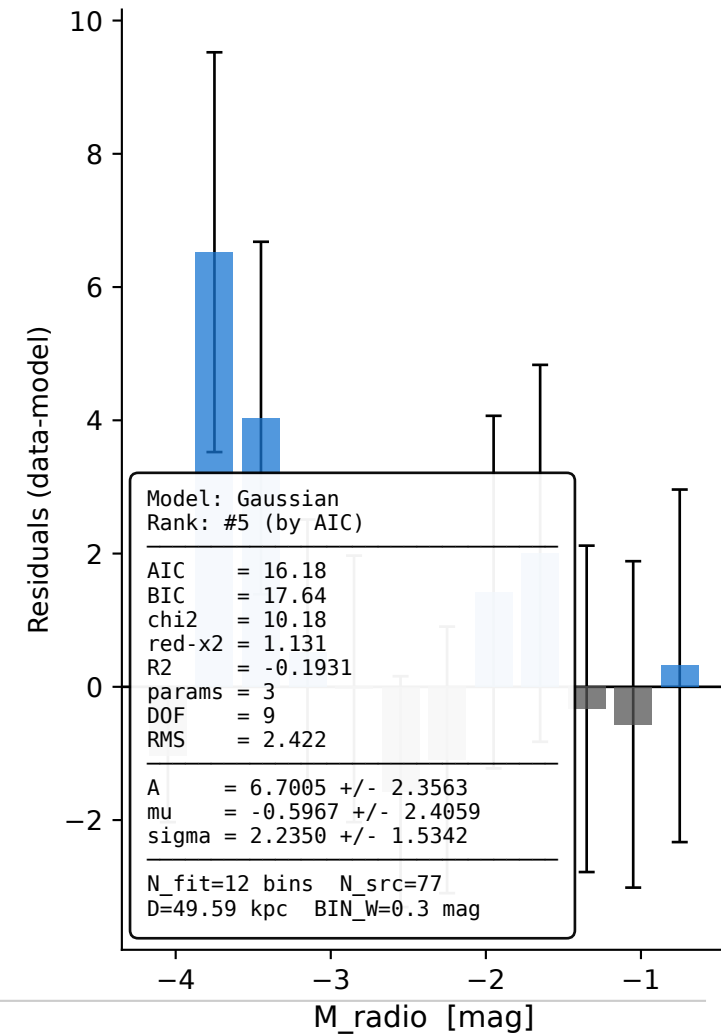
$$N(M) = A * \exp[-0.5 * ((M - \mu) / \sigma)^2]$$

A = amplitude,  $\mu$  = peak magnitude,  $\sigma$  = width

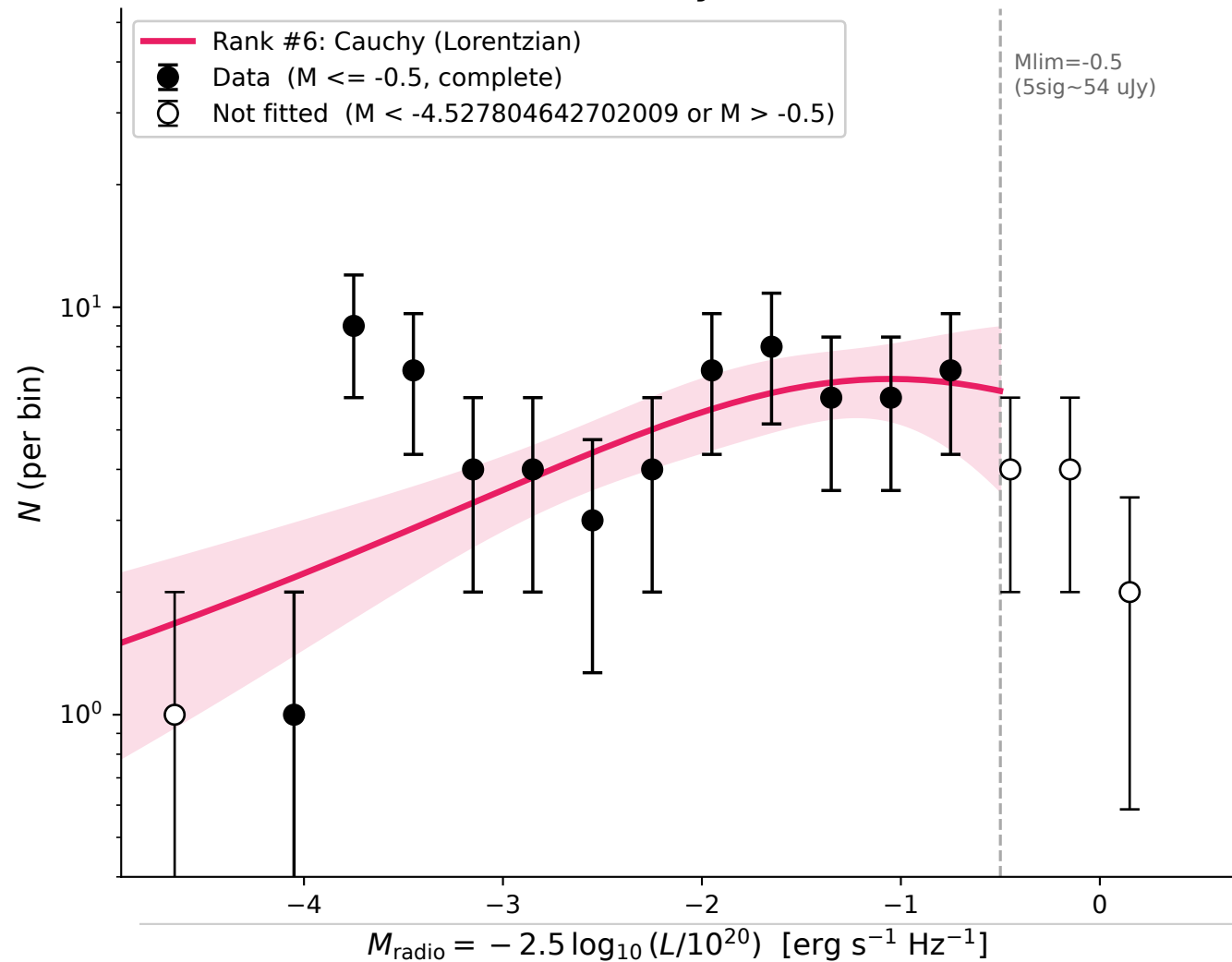
Fitted equation (parameter values substituted):

$$N(M) = 6.70 * \exp[-0.5 * ((M - (-0.597)) / 2.235)^2]$$

## Residuals



## Rank #6: Cauchy (Lorentzian)



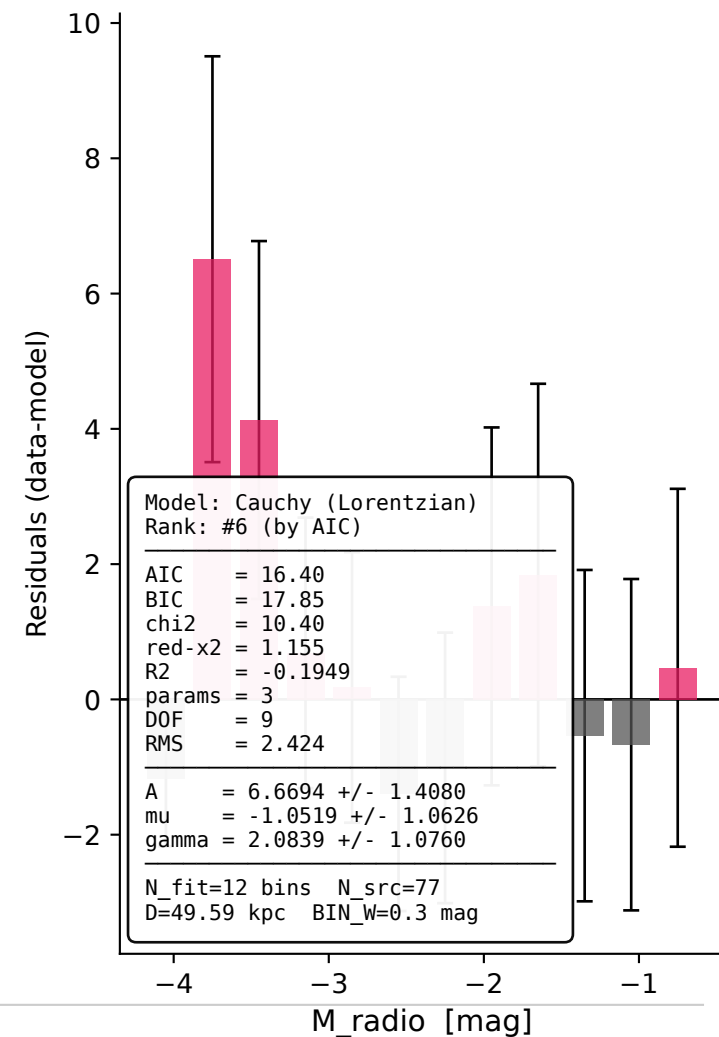
General model equation:

$$N(M) = A / [1 + ((M - \mu) / \gamma)^2]$$
 Heavy-tailed symmetric distribution (heavier tails than Gaussian)  
 A = peak height,  $\mu$  = centre,  $\gamma$  = half-width at half-maximum

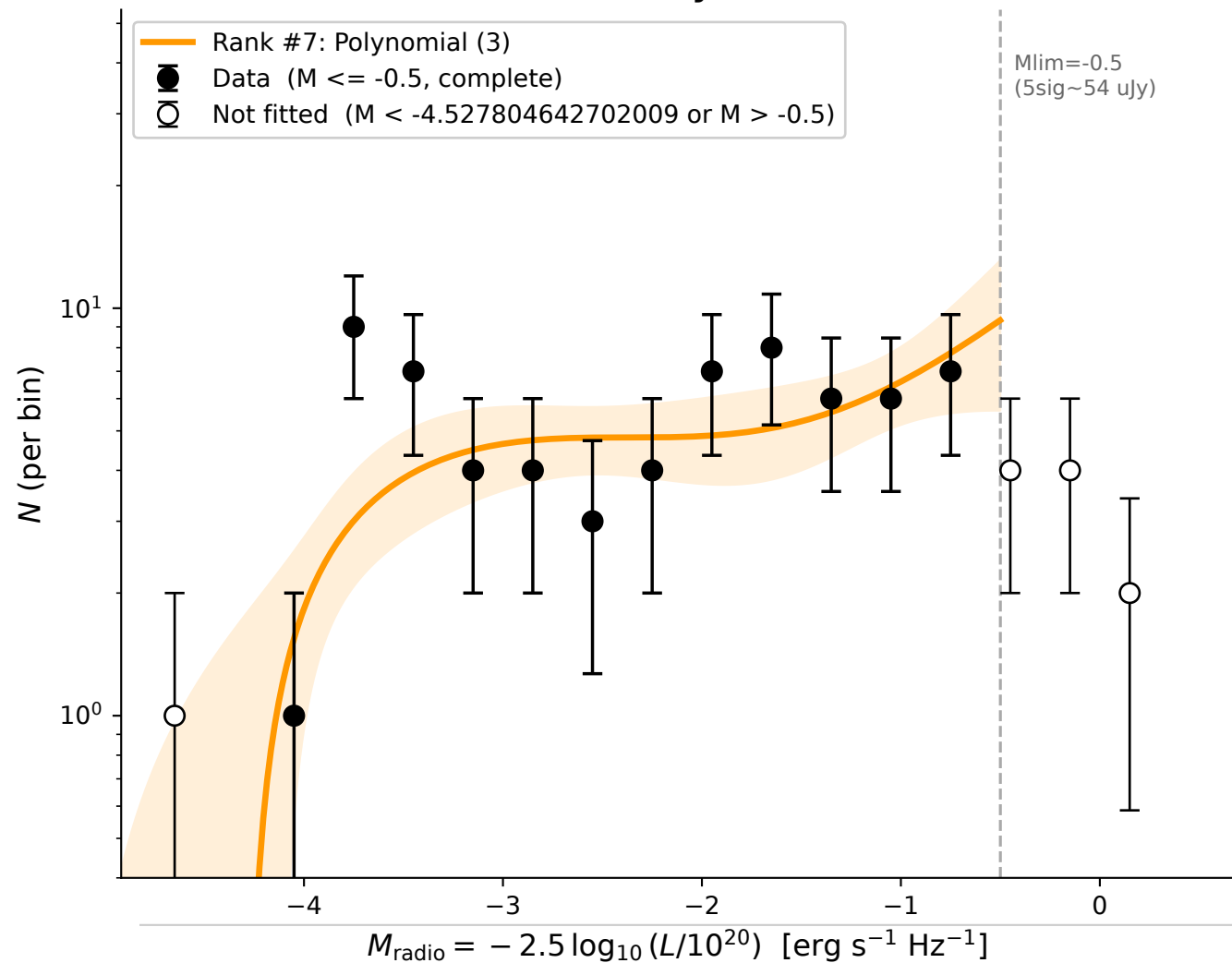
Fitted equation (parameter values substituted):

$$N(M) = 6.67 / [1 + ((M - (-1.052)) / 2.084)^2]$$

## Residuals



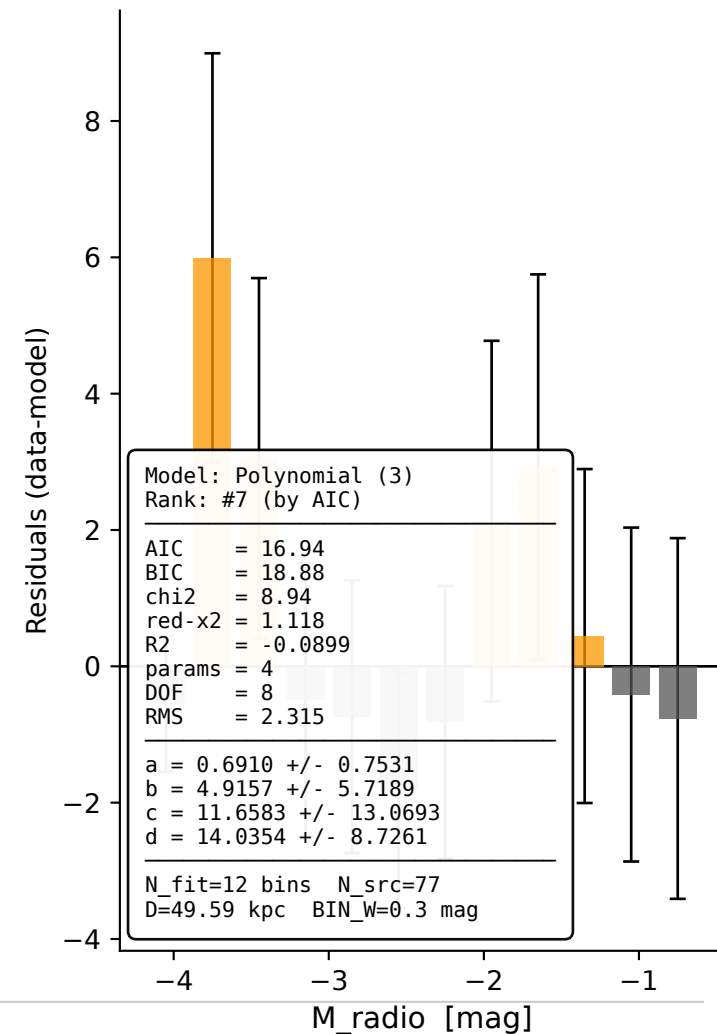
# Rank #7: Polynomial (3)



General model equation:

$$N(M) = a \cdot M^3 + b \cdot M^2 + c \cdot M + d$$

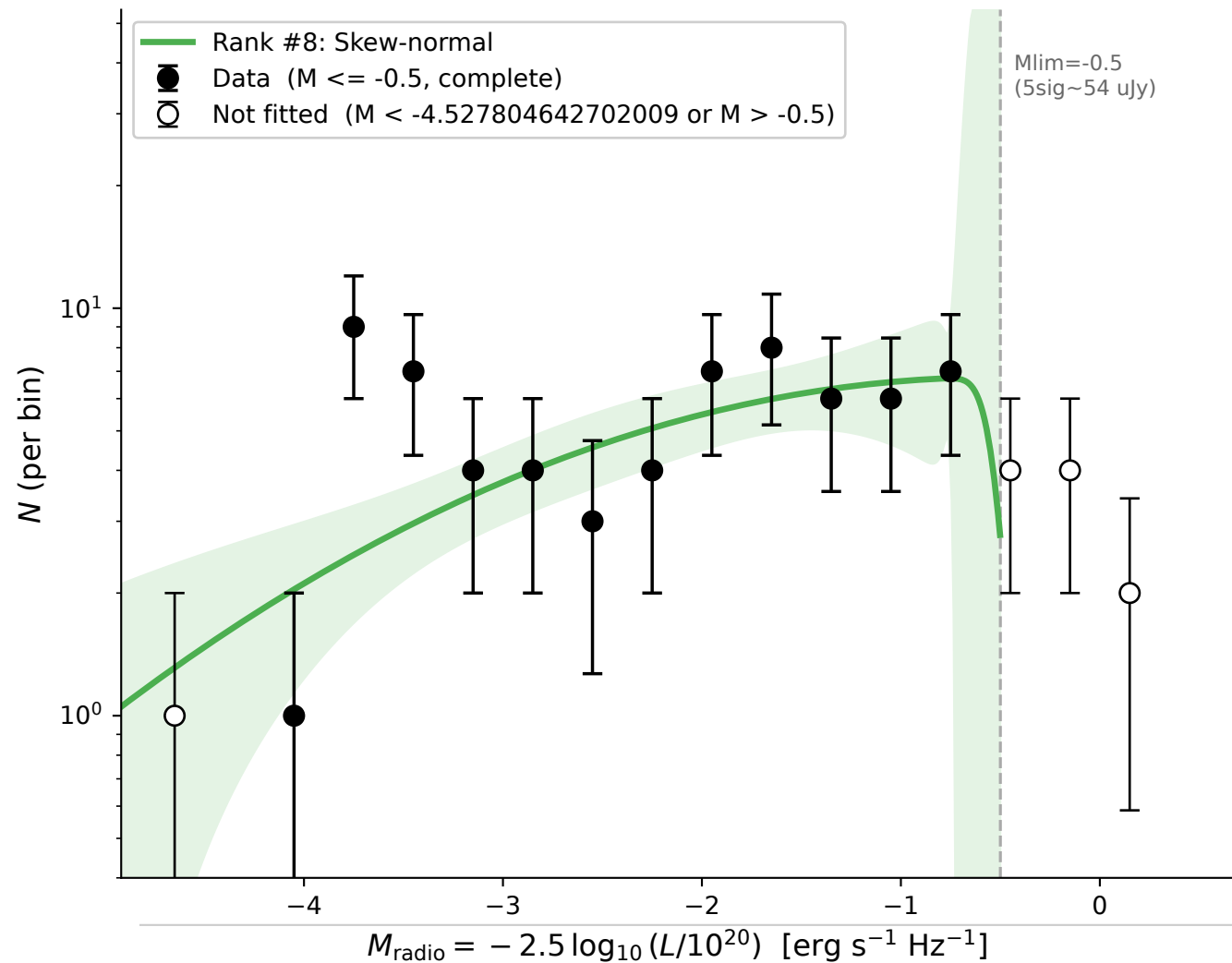
## Residuals



Fitted equation (parameter values substituted):

$$N(M) = 0.6910 \cdot M^3 + 4.9157 \cdot M^2 + 11.6583 \cdot M + 14.0354$$

## Rank #8: Skew-normal



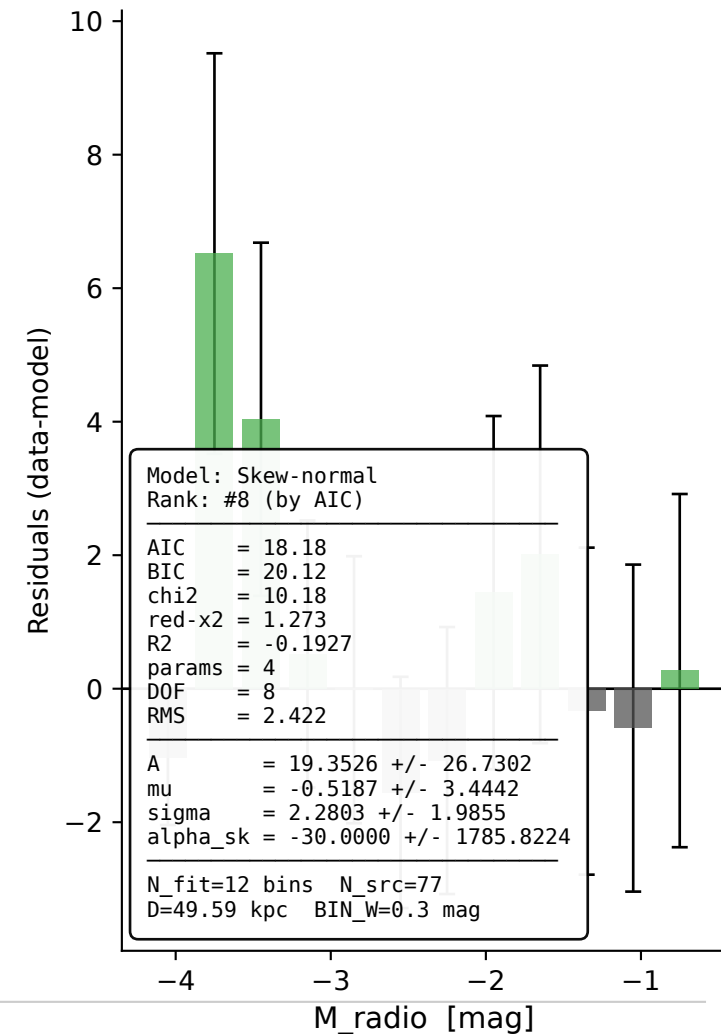
### General model equation:

$N(M) = A * f_{\text{SN}}(M; \mu, \sigma, \alpha_{\text{sk}})$   
 $f_{\text{SN}}(x) = (2/\sigma) * \phi((x-\mu)/\sigma) * \Phi(\alpha_{\text{sk}} * (x-\mu)/\sigma)$   
 $\phi$  = normal PDF,  $\Phi$  = normal CDF,  $\alpha_{\text{sk}} < 0$  = left-skewed

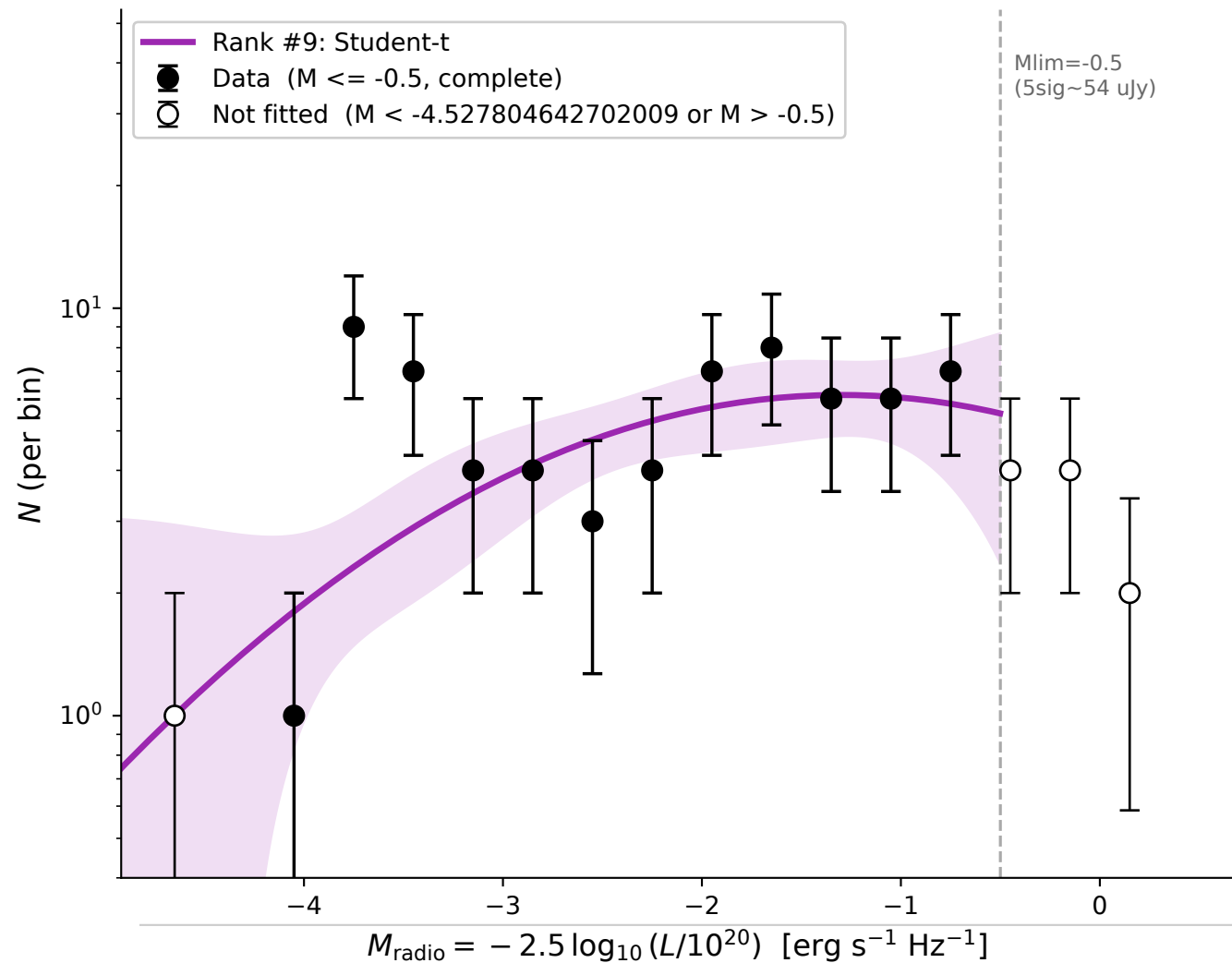
### Fitted equation (parameter values substituted):

$N(M) = 19.35 * f_{\text{SN}}(M; \mu=-0.519, \sigma=2.280, \alpha_{\text{sk}}=-30.000)$

## Residuals



## Rank #9: Student-t



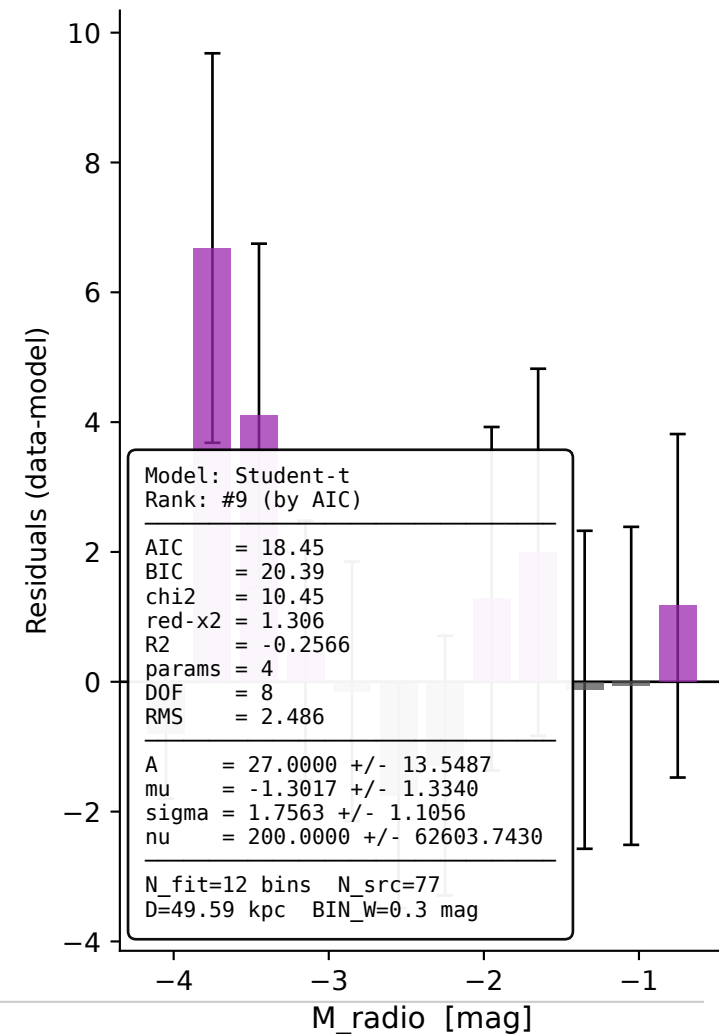
General model equation:

$N(M) = A * t_{\text{nu}}( (M - \mu) / \sigma )$   
 $t_{\text{nu}}(x)$  proportional to  $(1 + x^2/\nu)^{-(\nu+1)/2}$   
 $A$  = amplitude,  $\mu$  = centre,  $\sigma$  = scale,  $\nu$  = degrees of freedom

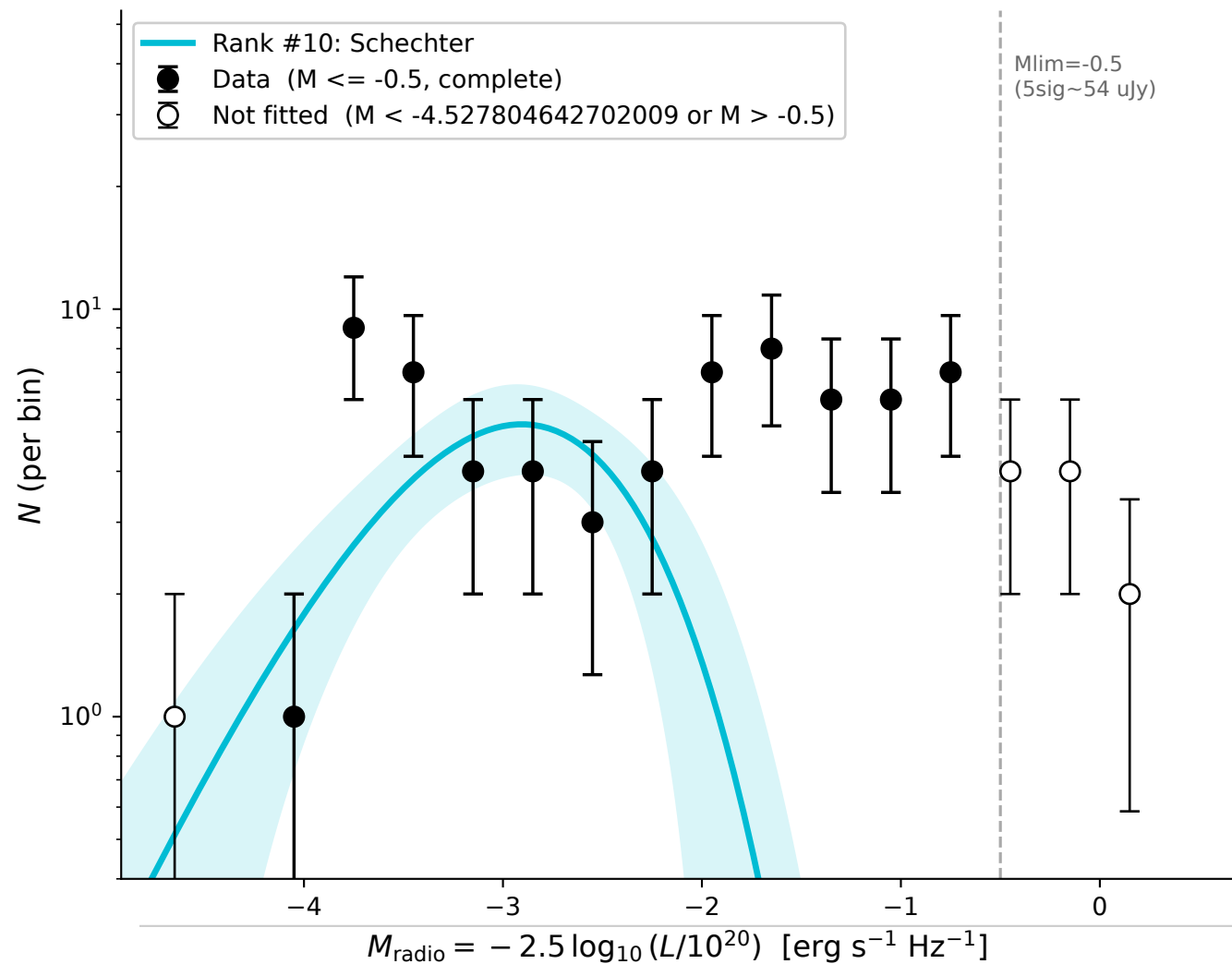
Fitted equation (parameter values substituted):

$N(M) = 27.00 * t_{200.0}( (M - (-1.302)) / 1.756 )$

## Residuals



## Rank #10: Schechter



General model equation:

$$N(M) = \phi * 10^{(0.4 * (\alpha + 1) * (M - M^*))} * \exp[-10^{(0.4 * (M - M^*))}]$$

$$\phi = \text{normalisation, } M^* = \text{bright cutoff, } \alpha = \text{faint-end slope}$$

Originally for galaxy luminosity functions (Schechter 1976)

Fitted equation (parameter values substituted):

$$N(M) = 4.458 * 10^{(0.4 * (1.871 + 1) * (M - (-4.050)))}$$

$$* \exp[-10^{(0.4 * (M - (-4.050)))}]$$

